

A BOUNDED FINITE-ONE DEGREE WHOSE ONE-ONE DEGREES FORM EXACTLY A DENSE LINEAR ORDER

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ABSTRACT. We construct a set $U \leq_T \emptyset''$ whose bounded finite-one degree, ordered internally by one-one reducibility, consists exactly of a countable dense linear order without endpoints. More precisely,

$$(\{[B]_1 : B \equiv_{\text{bfo}} U\}, \leq_1) \cong (\mathbb{Q}, \leq).$$

This exact realization contrasts with two previous results. In earlier work, a bounded finite-one degree was constructed that contains a copy of $(\mathbb{Q}, \leq)$, but the same degree also contains an infinite antichain of one-one degrees and, more generally, embedded copies of all countable partial orders; thus the dense chain does not exhaust the degree. In a different direction, m -rigidity yields an almost-sure and comeager obstruction: for a measure-1 and comeager class of sets, the corresponding bounded finite-one degree contains an infinite antichain of one-one degrees and hence is not linearly ordered. The present construction shows that, despite this typical negative behaviour, exact dense linear order can occur. In particular, it gives an affirmative answer to Open Question 3 of Richter, Stephan, and Zhang.

The proof has two main parts. A block homogenization construction produces a noncylindrical set U , a weak dyadic tower $(U_e)_{e \in \mathbb{N}}$, computable reservoirs of both colours, and a base absorption property. An abstract absorption-to-exhaustivity theorem then shows that every member of the bounded finite-one degree of U is one-one equivalent to some finite autojoin mU_e , thereby yielding the exhaustive classification above.

1. INTRODUCTION

We study the one-one degrees contained in a single bounded finite-one degree. A bounded finite-one reduction is a total computable many-one reduction whose fibres have a common finite bound. A bounded finite-one degree may therefore split into several one-one degrees, and one may ask which partially ordered structures arise when these degrees are ordered by one-one reducibility.

Richter, Stephan, and Zhang [6, Section 4] constructed a nonrecursive bounded finite-one degree whose one-one degrees form exactly a strictly ascending chain of order type ω . Their work thus shows that the full internal structure can be linearly ordered, rather than merely contain a linearly ordered subfamily. In the conclusion of their paper, they ask whether a dense linear order can also occur in this exact sense [6, Section 5, Open Question 3]:

Are there bounded finite-one degrees consisting exactly of a dense linearly ordered set of one-one degrees?

Two previous results put this question into perspective. The first gives a strong obstruction for typical sets. Recall that a set A is m -rigid if every total computable many-one self-reduction of A is eventually the identity. In [1, Section 6], it was shown that the bounded finite-one degree of every m -rigid set contains an infinite antichain of one-one degrees. Since the class of m -rigid sets has Lebesgue measure 1 and is comeager in Cantor space, it follows that, for almost every set A , and for a comeager class of sets A , the one-one degrees

Date: September 8, 2026.

2020 Mathematics Subject Classification. Primary 03D30; Secondary 06A05.

Key words and phrases. one-one reducibility, finite-one reducibility, bounded finite-one reducibility, degree structures, dense linear orders.

within $[A]_{\text{bfo}}$ are not linearly ordered at all. Richter, Stephan, and Zhang also observed the infinite-antichain phenomenon for Martin-Löf random sets [6, Section 4]; the m -rigidity argument extends it to a class that is also comeager. Thus the configuration sought in Open Question 3 is ruled out both almost everywhere and on a comeager class of sets.

The second result shows that density can nevertheless be realized as an embedded suborder. In [2], a block-density profile method produced a noncomputable set $A \leq_T \emptyset'$ whose bounded finite-one degree contains a copy of $(\mathbb{Q}, \leq)$. The same method also yielded a single bounded finite-one degree containing both such a dense chain and an infinite antichain, and, more generally, a degree into which every countable partial order embeds. In particular, the degree constructed there is not itself linearly ordered: the dense chain does not exhaust its one-one degrees. These results establish the presence of dense suborders, but do not settle the exact-realization problem. Nor do they imply that every bounded finite-one degree containing a dense chain must contain incomparable one-one degrees.

The present paper resolves this distinction by giving an affirmative answer to Open Question 3. We construct a bounded finite-one degree in which the dense chain exhausts the entire internal one-one degree structure.

Main Theorem (Corollary 5.1). *There exists a noncomputable set $U \leq_T \emptyset''$ such that*

$$(\{[B]_1 : B \equiv_{\text{bfo}} U\}, \leq_1) \cong (\mathbb{D}_{>0}, \leq) \cong (\mathbb{Q}, \leq),$$

where $\mathbb{D}_{>0} = \{m/2^n : m \geq 1, n \in \mathbb{N}\}$ is the set of positive dyadic rationals. In particular, the one-one degrees in $[U]_{\text{bfo}}$ form a countable dense linear order without endpoints.

This affirmative existence result contrasts with the almost-sure and comeager failure of the required configuration. Combining the theorem with [1, Section 6], the class of sets whose bounded finite-one degree has internal one-one order type $(\mathbb{Q}, \leq)$ is nonempty, but null and meager. Here measure and category refer to sets $A \in 2^{\mathbb{N}}$ representing the degrees, not to a measure or topology assumed on the quotient space of degrees.

More precisely, our construction supplies a sequence $U_0 = U, U_1, \dots$ with $U_n \equiv_1 2U_{n+1}$, where mA denotes the join of m copies of A . Every set $B \equiv_{\text{bfo}} U$ is one-one equivalent to some mU_n , and the comparison relation is given exactly by

$$mU_n \leq_1 kU_\ell \iff \frac{m}{2^n} \leq \frac{k}{2^\ell}.$$

Thus positive dyadic multiplicities provide a classification of all the one-one degrees in $[U]_{\text{bfo}}$, not merely parameters for a selected subfamily.

The proof is divided into two main parts. The Block Homogenization Construction, Theorem 3.3, builds a noncylindrical set U together with the sequence (U_n) , called a *weak dyadic tower*, and infinite computable subsets of both U_n and its complement at each level. The construction uses nested computable partitions into finite blocks. Homogenizing the number of points from the range of a computable injection that fall into selected blocks yields the *base absorption property*

$$X \leq_1 U \implies U \oplus X \equiv_1 mU_n \text{ for some } m \geq 1 \text{ and } n \in \mathbb{N}.$$

The Absorption-to-Exhaustivity Theorem, Theorem 4.6, then proves abstractly that these properties force every member of $[U]_{\text{bfo}}$ to have the required form. It combines a decomposition of bounded reductions into finitely many injective pieces with a finite-one cancellation lemma:

$$P \oplus Z \leq_1 Q \oplus Z \text{ and } Z \leq_{\text{fo}} Q \implies P \leq_1 Q.$$

Cancellation allows absorption to propagate along the tower and ultimately removes the common summand from the comparison that identifies an arbitrary $B \equiv_{\text{bfo}} U$ with a positive dyadic representative. This passage from absorption to an exhaustive classification is what excludes additional one-one degrees outside the dense chain.

Although $\emptyset''$ is used to select the construction data, every reduction in the proof is an ordinary computable function. No computable uniform family of reductions witnessing the tower equivalences is assumed: each individual reduction uses only finitely many fixed levels. The separation between the construction and the abstract classification makes explicit the two tasks involved in exact realization: producing the dyadic family and proving that it exhausts the degree.

2. PRELIMINARIES

We use standard terminology and notation from computability theory. General references are Rogers [5], Soare [7], and the two volumes of Odifreddi [3, 4].

All sets are subsets of $\mathbb{N} = \{0, 1, 2, \dots\}$ and are identified with their characteristic functions. We write $\bar{A} = \mathbb{N} \setminus A$ and refer to the values 0 and 1 of a characteristic function as the two membership colours. We write $\text{rng}(f)$ for the range of a function f .

We use “c.e.” for computably enumerable. Turing reducibility is denoted by $\leq_T$, and $\emptyset'$ and $\emptyset''$ denote the first and second Turing jumps of the empty set. The notation Σ_n^0 and Π_n^0 refers to the usual arithmetical hierarchy. Unless explicitly stated otherwise, all computability notions and all reductions are unrelativized.

A total computable function f is a many-one reduction from A to B if

$$x \in A \iff f(x) \in B$$

for every x . It is a *one-one reduction* if it is injective, a *finite-one reduction* if every fibre $f^{-1}(y)$ is finite, and a *bounded finite-one reduction* if there is a constant $k \geq 1$, possibly depending on f but independent of y , such that

$$|f^{-1}(y)| \leq k$$

for every y .

We write $\leq_1$, $\leq_{\text{fo}}$ and $\leq_{\text{bfo}}$ for the corresponding reducibilities, and use $\equiv_1$, $\equiv_{\text{fo}}$ and $\equiv_{\text{bfo}}$ for the induced equivalence relations. For $\rho \in \{1, \text{fo}, \text{bfo}\}$, let

$$[A]_\rho = \{B \subseteq \mathbb{N} : B \equiv_\rho A\}.$$

The equivalence classes induced by $\equiv_1$, $\equiv_{\text{fo}}$ and $\equiv_{\text{bfo}}$ are called one-one degrees, finite-one degrees and bounded finite-one degrees, respectively.

For every $m \geq 1$, fix a computable bijection

$$\langle \cdot, \cdot \rangle_m : \{0, \dots, m-1\} \times \mathbb{N} \longrightarrow \mathbb{N}$$

with computable inverse. We suppress the subscript m whenever the relevant finite product is clear from the context. We also fix a computable bijection $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ for the definition of cylinders.

For $m \geq 1$, the m -fold autojoin of A is

$$mA = \{\langle i, x \rangle : i < m \text{ \& } x \in A\}.$$

The binary join is

$$A \oplus B = \{\langle 0, x \rangle_2 : x \in A\} \cup \{\langle 1, x \rangle_2 : x \in B\}.$$

Up to a computable permutation,

$$mA \oplus nA = (m+n)A.$$

The cylinder over A is

$$\text{Cyl}(A) = \{\langle x, t \rangle : x \in A \text{ \& } t \in \mathbb{N}\}.$$

A set A is a *cylinder* if $A \equiv_1 \text{Cyl}(A)$.

We next introduce computable reservoirs, which allow computable join summands to be absorbed without changing the one-one degree. The idea is to map each reservoir injectively into part of itself, leaving infinitely many unused target positions of each membership colour

for the additional summand. Lemma 2.2 makes this precise and also shows that the reservoir property is preserved under finite autojoins.

Definition 2.1. A set A has *computable reservoirs of both colours* if there are computable infinite sets

$$R_1 \subseteq A \quad \text{and} \quad R_0 \subseteq \overline{A}.$$

Lemma 2.2 (Reservoir absorption). *Suppose that A has computable reservoirs of both colours. Then:*

(i) *for every computable set C ,*

$$A \oplus C \equiv_1 A;$$

(ii) *mA has computable reservoirs of both colours for every $m \geq 1$.*

Proof. Write

$$R_c = \{r_c(0) < r_c(1) < \dots\} \quad (c < 2).$$

Since each R_c is computable and infinite, its increasing enumeration r_c is total computable. Moreover, for $x \in R_c$, the unique index j such that $x = r_c(j)$ is computable from x , being equal to $|\{z < x : z \in R_c\}|$.

Define a computable colour-preserving injection $\rho : \mathbb{N} \rightarrow \mathbb{N}$ by

$$\rho(r_c(j)) = r_c(2j)$$

and $\rho(x) = x$ outside $R_0 \cup R_1$. Its range omits the two computable infinite sets

$$H_c = \{r_c(2j + 1) : j \in \mathbb{N}\}.$$

Indeed, membership in H_c is decidable: an element x belongs to H_c exactly when $x \in R_c$ and its index in the increasing enumeration of R_c is odd.

Since $H_c \subseteq R_c$ for each $c < 2$, we have

$$H_1 \subseteq R_1 \subseteq A, \quad H_0 \subseteq R_0 \subseteq \overline{A}.$$

Moreover,

$$\text{rng}(\rho) = \mathbb{N} \setminus (H_0 \cup H_1).$$

Thus H_1 and H_0 are disjoint computable infinite sets of free target positions of colours 1 and 0, respectively.

Map the A -summand of $A \oplus C$ by ρ . More precisely, define a map $\theta : \mathbb{N} \rightarrow \mathbb{N}$ on the two summands of $A \oplus C$ as follows. For every $x \in \mathbb{N}$, put

$$\theta(\langle 0, x \rangle) = \rho(x).$$

Map the C -summand into the free sets H_0 and H_1 according to membership colour. More precisely, for $c < 2$ and x such that $C(x) = c$, let

$$n_c(x) = |\{z < x : C(z) = c\}|$$

be the rank of x among the elements of C -colour c , and put

$$\theta(\langle 1, x \rangle) = r_c(2n_c(x) + 1).$$

Thus the copy of x in the C -summand is mapped to the element of H_c having index $n_c(x)$ in the enumeration

$$H_c = \{r_c(2j + 1) : j \in \mathbb{N}\}.$$

Since C is computable, the map

$$x \mapsto (C(x), n_{C(x)}(x))$$

is computable. Hence θ is total and computable. Its restriction to the first summand is injective because ρ is injective. Its restriction to the second summand is also injective: elements of the same C -colour have distinct ranks, while elements of different colours are mapped into the disjoint sets H_0 and H_1 .

Moreover,

$$\text{rng}(\rho) = \mathbb{N} \setminus (H_0 \cup H_1),$$

whereas the image of the second summand is contained in $H_0 \cup H_1$. Therefore the images of the two summands are disjoint, and θ is injective on all of $\mathbb{N}$.

Finally, θ preserves membership colour. On the first summand,

$$\langle 0, x \rangle \in A \oplus C \iff x \in A \iff \rho(x) \in A.$$

On the second summand,

$$\langle 1, x \rangle \in A \oplus C \iff x \in C \iff C(x) = 1 \iff r_{C(x)}(2n_{C(x)}(x) + 1) \in A,$$

because $H_1 \subseteq A$ and $H_0 \subseteq \overline{A}$. Consequently, θ is a one-one reduction of $A \oplus C$ to A . Conversely, the map

$$x \mapsto \langle 0, x \rangle$$

is a one-one reduction of A to $A \oplus C$. Hence

$$A \oplus C \equiv_1 A.$$

For (ii), fix $m \geq 1$ and, for each $c < 2$, put

$$R_c^{(m)} = \{\langle 0, x \rangle : x \in R_c\}.$$

Then $R_0^{(m)}$ and $R_1^{(m)}$ are computable and infinite. Moreover,

$$R_1^{(m)} \subseteq mA \quad \text{and} \quad R_0^{(m)} \subseteq \overline{mA}.$$

Thus mA has computable reservoirs of both colours.

□

The preceding absorption lemma will be used repeatedly in the following slightly more flexible form. If a computable part of B is a computably coordinatized copy of A , while the remainder of B is computable, then the remainder can be absorbed without changing the one-one degree.

Lemma 2.3 (Computable remainder absorption). *Let A have computable reservoirs of both colours. Let $B \subseteq \mathbb{N}$ and let $D \subseteq \mathbb{N}$ be computable. Suppose that there is a computable bijection $\pi : \mathbb{N} \rightarrow D$ satisfying*

$$x \in A \iff \pi(x) \in B,$$

and that $B \cap \overline{D}$ is computable. Then $A \equiv_1 B$.

Proof. The map π witnesses $A \leq_1 B$.

Put $K = B \cap \overline{D}$, which is computable by hypothesis. Since π is a computable bijection onto D , its inverse is computable on D . Define

$$f(y) = \begin{cases} \langle 0, \pi^{-1}(y) \rangle_2, & y \in D, \\ \langle 1, y \rangle_2, & y \notin D. \end{cases}$$

The function f is total computable because D is computable. It is injective on each of D and $\overline{D}$, and the images of these two sets lie in distinct columns.

If $y \in D$, then

$$y \in B \iff \pi^{-1}(y) \in A \iff f(y) \in A \oplus K.$$

If $y \notin D$, then

$$y \in B \iff y \in K \iff f(y) \in A \oplus K.$$

Thus $B \leq_1 A \oplus K$. By Lemma 2.2(i),

$$A \oplus K \equiv_1 A.$$

Consequently, $B \leq_1 A$, and hence $A \equiv_1 B$.

□

The construction and the later classification argument will be organised around two abstract properties. The first records a dyadic scaling between successive representatives, while the second says that adjoining any one-one subobject of the base can be absorbed into a finite autojoin at some level of the tower. We isolate these notions now so that the construction and the subsequent abstract argument can be kept separate.

Definition 2.4. A sequence $(U_e)_{e \in \mathbb{N}}$ is a *weak dyadic tower* over U_0 if

$$U_e \equiv_1 2U_{e+1} \quad (e \in \mathbb{N}).$$

No uniform procedure producing indices of these equivalences is assumed.

Definition 2.5. Let $(U_e)_{e \in \mathbb{N}}$ be a weak dyadic tower over $U = U_0$. The pair $(U, (U_e)_{e \in \mathbb{N}})$ has the *base absorption property* if, for every $X \leq_1 U$, there are $e \in \mathbb{N}$ and $m \geq 1$ such that

$$U \oplus X \equiv_1 mU_e.$$

3. THE BLOCK HOMOGENIZATION CONSTRUCTION

We now construct the set U and the weak dyadic tower that will satisfy the properties isolated above. The construction uses computable partitions into finite blocks and repeatedly reserves infinitely many blocks on which a prescribed computable injection has the same occupancy. The following constant-occupancy lemma is the combinatorial device that makes this homogenization possible.

Lemma 3.1 (Constant occupancy). *Let F be a computable equivalence relation all of whose classes have the same finite cardinality $d \geq 1$. Let*

$$C_0, C_1, C_2, \dots$$

be the canonical enumeration of its classes in increasing order of their least elements. Let $S \subseteq \mathbb{N}$ be computable and infinite, and let $\varphi : \mathbb{N} \rightarrow \mathbb{N}$ be a total computable injection. Then there is $k \in \{0, \dots, d\}$ such that, for every prescribed $r \geq 1$, there are pairwise disjoint computable infinite sets

$$P_0, \dots, P_{r-1} \subseteq S$$

such that

$$|\text{rng}(\varphi) \cap C_j| = k \quad (j \in P_0 \cup \dots \cup P_{r-1}).$$

The sets P_i may moreover be chosen to contain only indices of classes whose least element exceeds any prescribed constant.

Proof. Let $m_j = \min C_j$. Since F is computable, the set

$$M_F = \{m \in \mathbb{N} : (\forall z < m) \neg(z F m)\}$$

of least representatives of the F -classes is computable. Hence $j \mapsto m_j$, the increasing enumeration of M_F , is computable, and

$$x \in C_j \iff x F m_j.$$

Thus the family $(C_j)_{j \in \mathbb{N}}$ is uniformly computable. Moreover, since every class has cardinality d , the complete finite list of the elements of C_j can be computed uniformly from j , an index for F , and d . For $t \leq d$, put

$$V_t = \{j \in S : |\text{rng}(\varphi) \cap C_j| \geq t\}.$$

For $t = 0$, we have $V_0 = S$. For $1 \leq t \leq d$,

$$j \in V_t \iff j \in S \ \& \ \exists n_0 < \dots < n_{t-1} \bigwedge_{i < t} (\varphi(n_i) F m_j).$$

Since φ is total computable and injective, this is a c.e. condition, uniformly in t . Moreover,

$$V_0 = S \supseteq V_1 \supseteq \dots \supseteq V_d.$$

Let k be the largest $t \leq d$ for which V_t is infinite, and set $V_{d+1} = \emptyset$. Then V_{k+1} is finite. Since V_k is infinite, the set

$$E = V_k \setminus V_{k+1}$$

is infinite. Moreover, for every $j \in S$,

$$\begin{aligned} j \in E &\iff j \in V_k \text{ \& } j \notin V_{k+1} \\ &\iff |\text{rng}(\varphi) \cap C_j| \geq k \text{ \& } |\text{rng}(\varphi) \cap C_j| < k+1 \\ &\iff |\text{rng}(\varphi) \cap C_j| = k. \end{aligned}$$

Here, when $k = d$, we use the convention $V_{d+1} = \emptyset$ together with the bound

$$|\text{rng}(\varphi) \cap C_j| \leq d.$$

Thus E consists exactly of the indices $j \in S$ of classes having occupancy k .

The choice of k and the finite set V_{k+1} is nonuniform. Once that finite set has been fixed, however, E is c.e.: enumerate V_k while omitting the finitely many forbidden elements.

Fix $r \geq 1$ and a prescribed bound $b \in \mathbb{N}$. The set

$$E_b = \{j \in E : m_j > b\}$$

is c.e. and infinite, since $j \mapsto m_j$ is computable and strictly increasing, and only finitely many indices satisfy $m_j \leq b$.

Extract a total computable strictly increasing sequence

$$p(0) < p(1) < p(2) < \dots$$

of elements of E_b : choose an element when it is enumerated, and after choosing $p(n)$ wait for an enumerated element greater than $p(n)$ to define $p(n+1)$. Each search terminates because E_b is infinite.

For $i < r$, put

$$P_i = \{p(rn + i) : n \in \mathbb{N}\}.$$

For each $i < r$, the map $n \mapsto p(rn + i)$ is total computable and strictly increasing, so its range P_i is computable and infinite. The sets P_i are pairwise disjoint by injectivity of p and uniqueness of the remainder modulo r . Finally, every $j \in P_0 \cup \dots \cup P_{r-1}$ belongs to E_b , so its class has occupancy k and satisfies $m_j > b$. □

Remark 3.2 (Effectivity of the nonuniform choice). Given ordinary indices for F , S , and φ , together with d , the sets V_t are uniformly c.e. The property “ V_t is infinite” is Π_2^0 , so $\emptyset''$ can choose the largest suitable $k \in \{0, \dots, d\}$.

If $k = d$, then $V_{k+1} = \emptyset$ by convention. If $k < d$, the maximality of k implies that V_{k+1} is finite: otherwise $k+1 \leq d$ would be a larger index for which V_{k+1} is infinite. In this case, put $W = V_{k+1}$ and fix a computable increasing approximation $(W_s)_{s \in \mathbb{N}}$ by finite sets.

For each s , the condition

$$\forall u \geq s (W_u = W_s)$$

is Π_1^0 and hence decidable in $\emptyset'$. Since W is finite, a search using $\emptyset'$ finds such an s , and then $W = W_s$. Thus $\emptyset''$ suffices to obtain the exact finite content of V_{k+1} .

Once k and this finite set have been fixed, an ordinary c.e. index for E and, for any prescribed r and bound b , ordinary computable indices for the sets P_i can be obtained effectively from these data. The oracle is used only to select indices and finite constants that are subsequently hardcoded into ordinary programs; the resulting programs use no oracle.

We now assemble the preceding ingredients into the main construction. At each level, constant occupancy will provide the homogeneous block families needed for absorption, while separate permanent colour assignments diagonalize against possible cylinder embeddings. The same block structure will simultaneously generate the dyadic tower and preserve computable reservoirs of both colours.

Theorem 3.3 (Block Homogenization Construction). *There exist a set $U \leq_T \emptyset''$ and a sequence $(U_e)_{e \in \mathbb{N}}$ such that:*

- (i) $U_0 = U$, and (U_e) is a weak dyadic tower;
- (ii) U is noncomputable and is not a cylinder;
- (iii) every U_e has computable reservoirs of both colours;
- (iv) $(U, (U_e))$ has the base absorption property. More precisely, there is an $\emptyset''$ -computable list $(\varphi_e)_{e \in \mathbb{N}}$ consisting entirely of total computable injections from $\mathbb{N}$ to $\mathbb{N}$ and containing every such injection, possibly with repetitions, such that, for every $e \in \mathbb{N}$, there is $k_e \in \{0, \dots, 2^e\}$ satisfying

$$(3.1) \quad U \oplus \varphi_e^{-1}(U) \equiv_1 (2^e + k_e)U_e.$$

Proof. We use $\emptyset''$ only as a metatheoretic oracle. Let $(\psi_i)_{i \in \mathbb{N}}$ be a fixed standard enumeration of the partial computable functions from $\mathbb{N}$ to $\mathbb{N}$. Since the property

$$\psi_i \text{ is total and injective}$$

is Π_2^0 , the oracle $\emptyset''$ computes the increasing enumeration

$$a_0 < a_1 < a_2 < \dots$$

of all indices having this property. Put

$$\varphi_e = \psi_{a_e}.$$

Thus every φ_e is a total computable injection, and every total computable injection occurs in the list, possibly more than once extensionally.

Similarly, using a fixed standard enumeration of the partial computable functions from $\mathbb{N} \times \mathbb{N}$ to $\mathbb{N}$, fix an $\emptyset''$ -computable list

$$\gamma_0, \gamma_1, \dots$$

consisting entirely of total computable injections and containing every such injection. Repetitions are harmless.

We construct an increasing sequence

$$F_0 \subseteq F_1 \subseteq F_2 \subseteq \dots$$

of equivalence relations. Each F_e is an ordinary computable relation, although the sequence of their indices need not be computable. Every F_e -class will have cardinality 2^e . At the beginning of level e , each F_e -class carries an inherited computable label in $\{0, 1, *\}$; call this labelling λ_e^- . During level e we extend the commitments and obtain a completed computable labelling λ_e^+ . A class labelled 0 or 1 never changes colour, whereas a class labelled $*$ is uncommitted. At the end of each level there are infinitely many classes of each of the three labels.

Let

$$C_0^e, C_1^e, \dots$$

be the canonical enumeration of the F_e -classes by increasing least element. Since F_e is computable and every F_e -class has the known cardinality 2^e , the class index of a number, the complete list of the elements of a given class, and the least element of a class are uniformly computable from e and an index for F_e . We maintain the following invariant. At the beginning of level e , there are infinitely many F_e -classes labelled $*$, and every such class has least element greater than $e - 1$. Moreover, for $e > 0$ there are infinitely many classes carrying each of the labels 0 and 1. Classes carrying labels 0 or 1 have permanent final colour. At the end of level e , the uncommitted classes will all have least element greater than e .

At level 0, let F_0 be equality and let λ_0^- label every singleton by $*$. Suppose that F_e and λ_e^- have been defined.

Diagonalization against γ_e .

Choose the least F_e -class C , in the canonical enumeration $(C_j^e)_{j \in \mathbb{N}}$, such that C is labelled $*$ and $\min C > e$, and let

$$x = \min C.$$

Since the values

$$\gamma_e(x, 0), \gamma_e(x, 1), \dots$$

are pairwise distinct and C is finite, choose the least t such that

$$y = \gamma_e(x, t) \notin C.$$

If the F_e -class of y already has colour $c \in \{0, 1\}$, assign colour $1 - c$ to C . If the F_e -class of y is also labelled $*$, assign colour 0 to C and colour 1 to the F_e -class of y . These assignments are permanent.

Homogenization for φ_e . After the preceding finite action, let S_e be the set of indices j such that C_j^e is still labelled $*$ and has least element greater than e . This is a computable infinite set. Apply Lemma 3.1 with $d = 2^e$ and $r = 4$. We obtain a number $k_e \in \{0, \dots, 2^e\}$ and pairwise disjoint computable infinite sets

$$P_e, \quad Q_e^0, \quad Q_e^1, \quad R_e \subseteq S_e$$

such that

$$(3.2) \quad |\text{rng}(\varphi_e) \cap C_j^e| = k_e$$

for every j in their union. Put

$$(3.3) \quad H_e = P_e \cup Q_e^0 \cup Q_e^1.$$

Leave the classes indexed by P_e labelled $*$, assign colour 0 to the classes indexed by Q_e^0 , and assign colour 1 to the classes indexed by Q_e^1 . Assign colour 0 to every other class still labelled $*$, including those indexed by R_e . Classes already coloured by earlier levels or by the diagonalization retain their colours.

The resulting completed labelling is denoted by λ_e^+ . Thus the classes labelled $*$ by λ_e^+ are exactly those indexed by P_e , and in particular their least elements exceed e . There are infinitely many classes of all three labels: P_e supplies $*$, Q_e^0 supplies 0, and Q_e^1 supplies 1. The extra set R_e ensures that $\overline{H_e}$ is infinite.

Passage to F_{e+1} . For each label $c \in \{0, 1, *\}$, list in increasing order of their indices all F_e -classes on which λ_e^+ has value c , pair consecutive classes, and let each pair form one F_{e+1} -class. The new class inherits the common label c ; this defines λ_{e+1}^- . Then F_{e+1} is computable, each of its classes has size 2^{e+1} , and every F_{e+1} -class is the union of exactly two F_e -classes of the same completed label.

To see computability explicitly, given x, y , compute the indices of their F_e -classes and their λ_e^+ -labels. If the labels differ, then $x \not\equiv_{F_{e+1}} y$. If they agree, compute the ranks of the two class indices in the computable list of classes having that label; the two classes are amalgamated exactly when those ranks form the same consecutive pair. Since each label occurs on infinitely many F_e -classes, this pairing produces infinitely many F_{e+1} -classes of each label. Moreover, every F_{e+1} -class labelled $*$ has least element greater than e , because both of its F_e -children do. Thus the induction invariant holds at level $e + 1$. This completes the recursion.

Effectivity of the recursion. We verify by induction on e that $\emptyset''$ computes ordinary indices for F_e , λ_e^- , λ_e^+ , and the auxiliary computable sets chosen at level e . Given the data for level e , the diagonalization against γ_e is an ordinary computable search. The sets

$$V_t = \{j \in S_e : |\text{rng}(\varphi_e) \cap C_j^e| \geq t\} \quad (t \leq 2^e)$$

are uniformly c.e. The oracle $\emptyset''$ decides which of them are infinite and therefore chooses the maximal k_e . If $k_e = 2^e$, put $V_{k_e+1} = \emptyset$. Otherwise $\emptyset''$ finds the exact finite content of V_{k_e+1} .

With that finite set hardcoded, the proof of Lemma 3.1 yields ordinary computable indices for P_e, Q_e^0, Q_e^1, R_e . The completed labelling λ_e^+ and the pairing construction above then yield ordinary computable indices for F_{e+1} and λ_{e+1}^- . Thus the entire finite construction up to any prescribed level is uniformly recoverable from $\emptyset''$.

Definition of U . Every number eventually belongs to a coloured class. Indeed, after level e every uncommitted F_e -class has least element greater than e ; the class containing x has least element at most x , so it is coloured no later than level x . Define

$$(3.4) \quad x \in U \iff \text{the class containing } x \text{ is eventually assigned colour 1.}$$

Classes are amalgamated only with classes of the same label. Consequently U is saturated with respect to every F_e :

$$(3.5) \quad x F_e y \implies [x \in U \iff y \in U].$$

By the preceding effectivity verification, $\emptyset''$ computes the completed construction through level x . Since the colour of x is fixed by that level, it follows that $U \leq_T \emptyset''$.

Noncylindricity. Let $\hat{g} : \mathbb{N} \rightarrow \mathbb{N}$ be any total computable injection, and define

$$g(x, t) = \hat{g}(\langle x, t \rangle).$$

Then $g : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ is a total computable injection and hence occurs as some γ_e .

At level e we chose x, t and permanently assigned opposite colours to the F_e -classes of x and $g(x, t)$. Thus

$$U(x) \neq U(g(x, t)).$$

If $\hat{g}$ were a one-one reduction of $\text{Cyl}(U)$ to U , then

$$U(x) = \text{Cyl}(U)(\langle x, t \rangle) = U(\hat{g}(\langle x, t \rangle)) = U(g(x, t)),$$

contrary to the preceding inequality. Hence

$$(3.6) \quad \text{Cyl}(U) \not\leq_1 U.$$

The sets Q_0^1 and Q_0^0 are computable infinite subsets of U and $\overline{U}$, respectively. Thus U is infinite and coinfinite. If U were computable, then both colour classes of U and of $\text{Cyl}(U)$ would be computable and infinite. Matching, for each colour, the n -th element of the corresponding colour class of $\text{Cyl}(U)$ with the n -th element of that colour class of U would give a computable colour-preserving bijection, contradicting (3.6). Therefore U is noncomputable and noncylindrical.

The weak dyadic tower. Define

$$(3.7) \quad U_e = \{j : C_j^e \subseteq U\}.$$

Each F_{e+1} -class consists of exactly two F_e -classes of the same final colour. Given an F_e -class, one can computably find its F_{e+1} -parent and determine whether it is the first or second child; conversely, from a parent and a child number one can recover the child. This gives a computable colour-preserving bijection witnessing

$$(3.8) \quad U_e \equiv_1 2U_{e+1}.$$

Since F_0 is equality, $U_0 = U$. Moreover, the computable sets Q_e^1 and Q_e^0 are infinite subsets of U_e and $\overline{U_e}$, respectively. Thus every U_e has reservoirs of both colours.

For later use, let $d_e = 2^e$. The map sending x to the pair consisting of the rank of x in its F_e -class and the index of that class is a computable colour-preserving bijection. Hence

$$(3.9) \quad U \equiv_1 d_e U_e.$$

Base absorption. Fix e and put

$$X_e = \varphi_e^{-1}(U), \quad d = 2^e.$$

Let $h_e : \mathbb{N} \rightarrow H_e$ be the increasing enumeration of H_e , and set

$$(3.10) \quad V_e = \{s : C_{h_e(s)}^e \subseteq U\}.$$

The inverse images under h_e of Q_e^1 and Q_e^0 are computable infinite subsets of V_e and $\overline{V_e}$, respectively, so V_e has reservoirs of both colours. Since the classes labelled $*$ by λ_e^+ are exactly those indexed by $P_e \subseteq H_e$, every class whose index lies outside H_e already has its permanent final colour. Hence

$$U_e \cap \overline{H_e} = \{j \notin H_e : \lambda_e^+(j) = 1\}$$

is computable.

Apply Lemma 2.3 with

$$A = V_e, \quad B = U_e, \quad D = H_e, \quad \pi = h_e.$$

Since $s \in V_e$ iff $h_e(s) \in U_e$, the lemma gives

$$(3.11) \quad U_e \equiv_1 V_e.$$

Applying the witnessing one-one reductions separately on each of the d columns gives

$$dU_e \equiv_1 dV_e.$$

Together with (3.9),

$$(3.12) \quad U \equiv_1 dV_e.$$

Let

$$(3.13) \quad L_e = \bigcup_{j \in H_e} C_j^e \quad \text{and} \quad D_e = \varphi_e^{-1}(L_e).$$

Both sets are computable. If $k_e = 0$, then $D_e = \emptyset$ by (3.2), and membership in X_e is computable because every value of φ_e falls into an F_e -class outside H_e , whose colour is computable. Lemma 2.2 and (3.9) then give

$$U \oplus X_e \equiv_1 U \equiv_1 dU_e,$$

which is (3.1) in this case.

Suppose now that $k_e > 0$. For $x \in D_e$, let $j(x) \in H_e$ be the index of the F_e -class containing $\varphi_e(x)$, let

$$s(x) = h_e^{-1}(j(x)),$$

and define

$$(3.14) \quad r(x) = |\{z < x : \varphi_e(z) \in C_{j(x)}^e\}|.$$

By (3.2), $r(x) < k_e$. The map

$$(3.15) \quad x \mapsto (r(x), s(x))$$

is a computable bijection from D_e onto $\{0, \dots, k_e - 1\} \times \mathbb{N}$. Its inverse is computable: given (r, s) , search, in increasing order, for the $(r + 1)$ -st input whose φ_e -value belongs to $C_{h_e(s)}^e$; the search terminates because that class contains exactly k_e points of $\text{rng}(\varphi_e)$. Furthermore,

$$x \in X_e \iff C_{j(x)}^e \subseteq U \iff s(x) \in V_e.$$

Let $\beta_e : D_e \rightarrow \{0, \dots, k_e - 1\} \times \mathbb{N}$ denote the bijection in (3.15). Define $\pi_e : \mathbb{N} \rightarrow D_e$ by

$$\pi_e(\langle r, s \rangle_{k_e}) = \beta_e^{-1}(r, s) \quad (r < k_e, s \in \mathbb{N}).$$

Then π_e is a computable bijection.

The preceding colour calculation says exactly that

$$z \in k_e V_e \iff \pi_e(z) \in X_e.$$

On $\overline{D_e}$, membership in X_e is computable, because $\varphi_e(x)$ lies in an F_e -class outside H_e . Since $k_e V_e$ has reservoirs, Lemma 2.3, applied with

$$A = k_e V_e, \quad B = X_e, \quad D = D_e, \quad \pi = \pi_e,$$

yields

$$(3.16) \quad X_e \equiv_1 k_e V_e.$$

Using (3.12), (3.16), and (3.11), we obtain

$$U \oplus X_e \equiv_1 (d + k_e) V_e \equiv_1 (d + k_e) U_e.$$

This proves (3.1) for the fixed e .

All reductions used above are ordinary computable functions. For the fixed level e , their programs may hardcode the finite constant k_e and ordinary indices for the relevant computable level- e data, including F_e , H_e , λ_e^+ , φ_e , h_e , and the auxiliary computable maps. Later levels do not alter any of these data, do not split any F_e -class, and do not change any colour already assigned at level e . They may amalgamate the still uncommitted F_e -classes indexed by P_e , but their only effect relevant to the preceding colour calculations is to determine the final colours of those classes. These final colours are exactly recorded by the core set V_e . Hence all the reductions constructed above remain valid for the final sets.

Since every total computable injection occurs among the φ_e , the base absorption property follows. $\square$

Remark 3.4. Every fixed level yields ordinary computable witnesses. No computable uniform choice of witnesses throughout the whole infinite tower is assumed. Each witnessing reduction constructed above uses only finitely many fixed levels, whose ordinary computable indices may be hardcoded into its program.

The levels are fixed for the reduction as a whole, not merely finite in number on each individual input.

4. FROM ABSORPTION TO EXHAUSTIVITY

We now leave the specific block construction and turn to the abstract reduction-theoretic argument that converts base absorption into an exhaustive classification of the whole bounded finite-one degree. The main ingredients will be a cancellation principle for one-one reductions, the strict behaviour of finite autojoins of a noncylindrical set, and the resulting dyadic scalar calculus.

The first ingredient is a cancellation lemma. In general, a common summand cannot simply be cancelled from a one-one reduction. The additional hypothesis $Z \leq_{\text{fo}} Q$ provides enough finite control to make cancellation possible. The proof encodes the two reductions in a computable graph and then extracts, component by component, an injective reduction from P to Q .

Lemma 4.1 (Finite-one cancellation). *Let $P, Q, Z \subseteq \mathbb{N}$. If*

$$P \oplus Z \leq_1 Q \oplus Z \quad \text{and} \quad Z \leq_{\text{fo}} Q,$$

then

$$P \leq_1 Q.$$

Proof. Fix an injective reduction

$$f : P \oplus Z \leq_1 Q \oplus Z$$

and a finite-one reduction

$$g : Z \leq_{\text{fo}} Q.$$

Use four disjoint computable copies of $\mathbb{N}$, denoted by

$$P^-, \quad Z^-, \quad Q^+, \quad Z^+.$$

Write $x_P^-, z_Z^-, q_Q^+, w_Z^+$ for the vertices carrying the indicated underlying numbers, and give them membership colours $P(x)$, $Z(z)$, $Q(q)$, and $Z(w)$, respectively. We use the fixed binary coding of the two summands, so every value of f has a unique form $\langle j, y \rangle$ with $j < 2$.

Build a computable, and hence c.e., undirected graph as follows. For each x , if

$$f(\langle 0, x \rangle) = \langle 0, q \rangle,$$

add an edge from x_P^- to q_Q^+ , while if

$$f(\langle 0, x \rangle) = \langle 1, z \rangle,$$

add an edge from x_P^- to z_Z^+ . Similarly, for each z , use the value of $f(\langle 1, z \rangle)$ to join z_Z^- to the corresponding vertex in $Q^+ \cup Z^+$. In addition, add

- (1) an edge between z_Z^- and z_Z^+ for every z ;
- (2) an edge between z_Z^+ and $g(z)_Q^+$ for every z .

Adjacency is decidable from the types and underlying numbers of the two vertices, by evaluating f or g and checking the matching condition where appropriate. No membership test in P , Q , or Z is used in constructing the graph. Every edge joins vertices of the same membership colour: this follows from the reduction property of f for the first family of edges, from the identical Z -labels for the matching edges, and from the reduction property of g for the last family. Hence each connected component is monochromatic.

Let K be a finite component. The matching between Z^- and Z^+ gives

$$|K \cap Z^-| = |K \cap Z^+|.$$

Since f is injective and maps every left vertex of K to a right vertex of K ,

$$|K \cap P^-| + |K \cap Z^-| \leq |K \cap Q^+| + |K \cap Z^+|.$$

Hence

$$(4.1) \quad |K \cap P^-| \leq |K \cap Q^+|.$$

Every infinite component contains infinitely many Q^+ -vertices. Indeed, let K be a component such that $K \cap Q^+$ is finite. If $z_Z^+ \in K$, then the edge from z_Z^+ to $g(z)_Q^+$ shows that $g(z)_Q^+ \in K$. Hence

$$\{z : z_Z^+ \in K\} \subseteq \bigcup_{\{q : q_Q^+ \in K\}} g^{-1}(q).$$

The set on the right is finite, since $K \cap Q^+$ is finite and g is finite-one. Thus $K \cap Z^+$ is finite. The matching edges then show that $K \cap Z^-$ is finite. Finally, f maps the left-hand vertices of K injectively into the finite set

$$K \cap (Q^+ \cup Z^+),$$

so $K \cap P^-$ is finite as well. Therefore K is finite. This proves the claim.

Construct $h : \mathbb{N} \rightarrow \mathbb{N}$ recursively in increasing order of its argument. The connected component of a vertex is uniformly c.e., since it can be enumerated by enumerating all finite paths from that vertex.

Suppose that $h(0), \dots, h(p-1)$ have been defined. Enumerate the component of p_P^- until a vertex q_Q^+ is found such that

$$q \notin \{h(0), \dots, h(p-1)\},$$

and put $h(p) = q$.

This procedure does not require deciding whether the component is finite or infinite, nor computing the complete content of any fibre of g . It uses only the enumeration of the component and the finite list of previously chosen values. This search terminates. If the component is infinite, it contains infinitely many Q^+ -vertices, whereas only finitely many values have previously been used. Suppose instead that the component is a finite component

K . Any previously used Q^+ -vertex belonging to K was chosen for an earlier argument p' with $p'_P \in K$. Since the current vertex p_P^- also lies in K , at most

$$|K \cap P^-| - 1$$

vertices of $K \cap Q^+$ have previously been used. By (4.1),

$$|K \cap P^-| \leq |K \cap Q^+|,$$

so an unused Q^+ -vertex remains.

It follows that h is total and computable. It is injective by construction. Since p_P^- and $h(p)_Q^+$ belong to the same monochromatic component,

$$p \in P \iff h(p) \in Q.$$

Thus $P \leq_1 Q$. □

Cancellation allows us to convert noncylindricity into a strict separation between finite autojoins. The next lemma shows that, for a noncylindrical set, adding further copies can never collapse the one-one degree.

Lemma 4.2 (Strict finite autojoins). *If A is not a cylinder, then*

$$A <_1 2A <_1 3A <_1 \dots$$

Proof. First note that $1A \equiv_1 A$, via the computable bijection $x \mapsto \langle 0, x \rangle_1$.

For every $m \geq 1$, the map

$$\langle i, x \rangle_m \mapsto \langle i, x \rangle_{m+1}$$

is a one-one reduction of mA to $(m+1)A$.

Suppose first that

$$2A \leq_1 A,$$

and let f be an injective reduction witnessing this. For $i < 2$, define

$$a_i(x) = f(\langle i, x \rangle_2).$$

Then $a_0, a_1 : \mathbb{N} \rightarrow \mathbb{N}$ are total computable colour-preserving injections. Their ranges are disjoint, because f is injective and the two columns of $2A$ are disjoint. For each $n \in \mathbb{N}$, put

$$e_n = a_1^n \circ a_0,$$

where $a_1^0 = \text{id}_{\mathbb{N}}$.

Each e_n is a total computable colour-preserving injection, uniformly in n . The ranges of the maps e_n are pairwise disjoint. Indeed, suppose that $m < n$ and

$$e_m(x) = e_n(y).$$

Then

$$a_1^m(a_0(x)) = a_1^m(a_1^{n-m}(a_0(y))).$$

Since a_1^m is injective,

$$a_0(x) = a_1^{n-m}(a_0(y)).$$

The left-hand side belongs to $\text{rng}(a_0)$, while the right-hand side belongs to $\text{rng}(a_1)$ because $n - m \geq 1$. This contradicts the disjointness of the two ranges.

It follows that the map

$$\langle x, n \rangle \mapsto e_n(x)$$

is a total computable injection. Moreover,

$$\langle x, n \rangle \in \text{Cyl}(A) \iff x \in A \iff e_n(x) \in A,$$

so it is a one-one reduction of $\text{Cyl}(A)$ to A . Conversely, the map

$$x \mapsto \langle x, 0 \rangle$$

is always a one-one reduction of A to $\text{Cyl}(A)$. Hence

$$A \equiv_1 \text{Cyl}(A),$$

contrary to the hypothesis.

Now suppose that

$$(m+1)A \leq_1 mA$$

for some $m \geq 2$. Put $Z = (m-1)A$. Up to computable permutations of the columns,

$$(m+1)A \equiv_1 2A \oplus Z, \quad mA \equiv_1 A \oplus Z.$$

Thus the assumed reduction yields

$$2A \oplus Z \leq_1 A \oplus Z.$$

The projection

$$\langle i, x \rangle_{m-1} \mapsto x$$

is a bounded finite-one reduction of Z to A , and in particular

$$Z \leq_{\text{fo}} A.$$

Lemma 4.1, applied with

$$P = 2A, \quad Q = A, \quad Z = (m-1)A,$$

therefore gives

$$2A \leq_1 A,$$

which, by the first part of the proof, would make A a cylinder. This is again a contradiction.

Consequently,

$$(m+1)A \not\leq_1 mA$$

for every $m \geq 1$. Since $mA \leq_1 (m+1)A$, we conclude that

$$A <_1 2A <_1 3A <_1 \dots$$

□

Lemma 4.3 (Transfer of reservoirs). *If A has computable reservoirs of both colours and $A \leq_{\text{fo}} B$, then B has computable reservoirs of both colours.*

Proof. Let

$$R_1 \subseteq A \quad \text{and} \quad R_0 \subseteq \overline{A}$$

be computable infinite reservoirs, and let

$$f : A \leq_{\text{fo}} B.$$

For $c < 2$, put

$$W_c = f[R_c].$$

Since R_c is computable and f is total computable, W_c is c.e. Moreover, the reduction property of f gives

$$W_1 \subseteq B \quad \text{and} \quad W_0 \subseteq \overline{B}.$$

Each W_c is infinite. Indeed, if W_c were finite, then

$$R_c \subseteq \bigcup_{y \in W_c} f^{-1}(y),$$

which would be finite, since W_c is finite and every fibre of f is finite. This contradicts the infinitude of R_c .

Every infinite c.e. set contains an infinite computable subset. For completeness, fix $c < 2$ and an enumeration of W_c . Choose $w_c(0)$ by waiting for an element to be enumerated. After $w_c(n)$ has been chosen, wait for an enumerated element greater than $w_c(n)$ and choose it as $w_c(n+1)$. Each search terminates because W_c is infinite. Thus

$$w_c(0) < w_c(1) < \dots$$

is a total computable sequence of elements of W_c .

Its range

$$S_c = \{w_c(n) : n \in \mathbb{N}\}$$

is infinite and computable. Indeed, strict increase implies $w_c(n) \geq n$, so

$$x \in S_c \iff (\exists n \leq x) [w_c(n) = x],$$

which is a decidable condition. By construction, $S_c \subseteq W_c$.

Consequently,

$$S_1 \subseteq B \quad \text{and} \quad S_0 \subseteq \overline{B},$$

so B has computable reservoirs of both colours. $\square$

We next record the coarse equivalence shared by all finite multiples along a weak dyadic tower. Although these sets will represent distinct one-one degrees once noncylindricity is imposed, they all remain inside the same bounded finite-one degree.

Lemma 4.4 (Bounded equivalence of nonzero tower multiples). *Let (U_n) be a weak dyadic tower over $U = U_0$. Then, for every $n \in \mathbb{N}$ and every $m \geq 1$,*

$$mU_n \equiv_{\text{bfo}} U.$$

Proof. First observe that one-one equivalence is preserved under finite autojoins. Indeed, if $f : A \leq_1 B$ and $r \geq 1$, then

$$\langle i, x \rangle_r \mapsto \langle i, f(x) \rangle_r$$

is a one-one reduction of rA to rB .

We claim that, for every $n \in \mathbb{N}$,

$$U \equiv_1 2^n U_n.$$

For $n = 0$, this follows from $U_0 = U$ and the computable bijection $x \mapsto \langle 0, x \rangle_1$ witnessing $U_0 \equiv_1 1U_0$. Suppose that

$$U \equiv_1 2^n U_n.$$

Since $U_n \equiv_1 2U_{n+1}$, applying the witnessing reductions on each of the 2^n columns gives

$$2^n U_n \equiv_1 2^n (2U_{n+1}) \equiv_1 2^{n+1} U_{n+1},$$

where the last equivalence is induced by a computable permutation of the columns. This proves the claim by induction. Notice that only finitely many tower equivalences are used for each fixed n , so no uniformity in the tower is required.

Now fix $r \geq 1$. The map

$$x \mapsto \langle 0, x \rangle_r$$

is a one-one reduction of U_n to rU_n . Conversely, under the fixed computable coding of $\{0, \dots, r-1\} \times \mathbb{N}$, the projection

$$\langle i, x \rangle_r \mapsto x$$

is a bounded finite-one reduction of rU_n to U_n , with every fibre having cardinality exactly r . Hence

$$rU_n \equiv_{\text{bfo}} U_n \quad (r \geq 1).$$

Taking $r = 2^n$ and using

$$U \equiv_1 2^n U_n,$$

we obtain

$$U \equiv_{\text{bfo}} U_n.$$

Taking instead $r = m$ gives

$$mU_n \equiv_{\text{bfo}} U_n.$$

By transitivity of bounded finite-one reducibility,

$$mU_n \equiv_{\text{bfo}} U,$$

as required. $\square$

The dyadic tower now admits a natural numerical parametrization. Passing from U_n to U_{n+1} divides the corresponding scale by two, while taking a finite autojoin multiplies it by an integer. Noncylindricity ensures that these numerical values faithfully record one-one reducibility, and binary join will correspond to addition.

Let

$$\mathbb{D}_{>0} = \left\{ \frac{m}{2^n} : m \geq 1, n \in \mathbb{N} \right\}$$

be the set of positive dyadic rationals.

Lemma 4.5 (Scalar calculus). *Let (U_n) be a weak dyadic tower over a noncylindrical set $U = U_0$. For each $q \in \mathbb{D}_{>0}$, choose a representation $q = m/2^n$ with $m \geq 1$ and $n \in \mathbb{N}$, and put*

$$S_q = mU_n.$$

Then the one-one degree of S_q is independent of the chosen representation, and, for all $q, r \in \mathbb{D}_{>0}$,

$$(4.2) \quad S_q \leq_1 S_r \iff q \leq r,$$

$$(4.3) \quad S_q \oplus S_r \equiv_1 S_{q+r}.$$

Proof. First observe that finite autojoins preserve one-one equivalence. Indeed, if $f : A \leq_1 B$ and $a \geq 1$, then

$$\langle i, x \rangle_a \mapsto \langle i, f(x) \rangle_a$$

is a one-one reduction of aA to aB .

Consequently, if $N \geq n$, then iterating the finitely many tower equivalences between levels n and N gives

$$mU_n \equiv_1 m2^{N-n}U_N.$$

Only finitely many witnesses are used for each fixed n and N , so no uniformity in the tower is required.

We next show that every U_N is noncylindrical. Cylinderhood is invariant under one-one equivalence. Indeed, if $A \equiv_1 B$ and $\text{Cyl}(B) \leq_1 B$, then a one-one reduction $f : A \leq_1 B$ induces

$$\text{Cyl}(A) \leq_1 \text{Cyl}(B)$$

by

$$\langle x, t \rangle \mapsto \langle f(x), t \rangle.$$

Hence

$$\text{Cyl}(A) \leq_1 \text{Cyl}(B) \leq_1 B \leq_1 A.$$

Since the canonical map

$$x \mapsto \langle x, 0 \rangle$$

always gives $A \leq_1 \text{Cyl}(A)$, the set A is a cylinder.

Suppose now that U_N were a cylinder. For every $a \geq 1$, the map

$$\langle i, x \rangle_a \mapsto \langle x, i \rangle$$

is a one-one reduction of aU_N to $\text{Cyl}(U_N)$. Thus

$$aU_N \leq_1 \text{Cyl}(U_N) \leq_1 U_N,$$

while the inclusion into one column gives

$$U_N \leq_1 aU_N.$$

Therefore $aU_N \equiv_1 U_N$ for every $a \geq 1$. In particular,

$$2^N U_N \equiv_1 U_N.$$

Since the tower gives

$$U \equiv_1 2^N U_N,$$

we would have $U \equiv_1 U_N$, and the invariance of cylinderhood would make U a cylinder, a contradiction. Hence every U_N is noncylindrical.

We now prove that the degree of S_q is independent of the chosen representation of q . Suppose

$$q = \frac{m}{2^n} = \frac{m'}{2^{n'}}.$$

Choose $N \geq n, n'$. Then

$$m2^{N-n} = m'2^{N-n'},$$

and hence

$$mU_n \equiv_1 m2^{N-n}U_N = m'2^{N-n'}U_N \equiv_1 m'U_{n'}.$$

Thus the one-one degree of S_q depends only on q .

Next let

$$q = \frac{m}{2^n}, \quad r = \frac{k}{2^\ell},$$

choose $N \geq n, \ell$, and put

$$a = m2^{N-n}, \quad b = k2^{N-\ell}.$$

Then

$$(4.4) \quad S_q \equiv_1 aU_N, \quad S_r \equiv_1 bU_N.$$

Moreover,

$$q \leq r \iff a \leq b.$$

Since U_N is noncylindrical, Lemma 4.2 implies

$$aU_N \leq_1 bU_N \iff a \leq b.$$

Indeed, if $a \leq b$, inclusion of columns gives $aU_N \leq_1 bU_N$.

If $a > b$ and $aU_N \leq_1 bU_N$, then

$$(b+1)U_N \leq_1 aU_N \leq_1 bU_N,$$

contradicting

$$bU_N <_1 (b+1)U_N.$$

Together with (4.4), this proves

$$S_q \leq_1 S_r \iff q \leq r.$$

One-one equivalence is also preserved under binary joins, by applying the corresponding reductions separately on the two columns. Finally,

$$\begin{aligned} S_q \oplus S_r &\equiv_1 aU_N \oplus bU_N \\ &\equiv_1 (a+b)U_N, \end{aligned}$$

where the last equivalence is induced by a computable permutation of the columns. Since

$$q + r = \frac{a+b}{2^N},$$

the well-definedness already proved gives

$$(a+b)U_N \equiv_1 S_{q+r}.$$

Therefore

$$S_q \oplus S_r \equiv_1 S_{q+r}.$$

□

We can now combine the preceding ingredients. The base absorption property initially controls only joins $U \oplus X$ with $X \leq_1 U$; cancellation will propagate this control throughout the dyadic tower. An arbitrary bounded finite-one reduction can then be decomposed into finitely many injective pieces, allowing the scalar calculus to be applied successively. The resulting scalar increment will finally identify the one-one degree of the original set.

Theorem 4.6 (Absorption-to-Exhaustivity). *Let $U = U_0, U_1, \dots$ satisfy the following conditions:*

- (i) (U_n) is a weak dyadic tower;
- (ii) U is not a cylinder;
- (iii) U has computable reservoirs of both colours;
- (iv) $(U, (U_n))$ has the base absorption property.

Then every $B \equiv_{\text{bfo}} U$ is one-one equivalent to mU_n for some $m \geq 1$ and $n \in \mathbb{N}$. Consequently,

$$(4.5) \quad (\{[B]_1 : B \equiv_{\text{bfo}} U\}, \leq_1) \cong (\mathbb{D}_{>0}, \leq).$$

Proof. We divide the proof into four steps. Throughout the proof, saying that a set is a scalar means that it is one-one equivalent to S_q for some $q \in \mathbb{D}_{>0}$. Recall that

$$U \equiv_1 S_1 \quad \text{and} \quad U_n \equiv_1 S_{2^{-n}} \quad (n \in \mathbb{N}).$$

Step 1: absorption propagates from U to every scalar. Let $X \leq_1 U$. By base absorption,

$$(4.6) \quad U \oplus X \equiv_1 S_q$$

for some $q \in \mathbb{D}_{>0}$. Since $U \leq_1 U \oplus X$, scalar calculus gives $q \geq 1$.

Fix $n > 0$ and put

$$Z_n = (2^n - 1)U_n.$$

Then

$$U \equiv_1 Z_n \oplus U_n,$$

and Z_n has scalar value

$$z_n = 1 - 2^{-n}.$$

From (4.6),

$$(4.7) \quad Z_n \oplus (U_n \oplus X) \equiv_1 S_q \equiv_1 Z_n \oplus S_{q-z_n}.$$

Here $q - z_n \in \mathbb{D}_{>0}$, since $q \geq 1$ and $z_n = 1 - 2^{-n}$. After pre- and post-composing the forward reduction in (4.7) with the canonical computable permutations witnessing associativity and commutativity of finite joins, we obtain

$$(U_n \oplus X) \oplus Z_n \leq_1 S_{q-z_n} \oplus Z_n.$$

Apply Lemma 4.1 with

$$P = U_n \oplus X, \quad Q = S_{q-z_n}, \quad Z = Z_n.$$

Since $n > 0$, the set

$$Z_n = (2^n - 1)U_n$$

is a nonzero scalar, and since $q - z_n > 0$, the set S_{q-z_n} is also a nonzero scalar. By Lemma 4.4,

$$Z_n \leq_{\text{bfo}} U \leq_{\text{bfo}} S_{q-z_n}.$$

By transitivity,

$$Z_n \leq_{\text{bfo}} S_{q-z_n},$$

and hence

$$Z_n \leq_{\text{fo}} S_{q-z_n}.$$

Lemma 4.1 therefore gives

$$U_n \oplus X \leq_1 S_{q-z_n}.$$

For the reverse direction, after applying the corresponding computable permutations to the reverse reduction in (4.7), we obtain

$$S_{q-z_n} \oplus Z_n \leq_1 (U_n \oplus X) \oplus Z_n.$$

Apply Lemma 4.1 with

$$P = S_{q-z_n}, \quad Q = U_n \oplus X, \quad Z = Z_n.$$

The projection gives

$$Z_n \leq_{\text{fo}} U_n,$$

and the canonical inclusion gives

$$U_n \leq_1 U_n \oplus X.$$

Hence, by transitivity,

$$Z_n \leq_{\text{fo}} U_n \oplus X.$$

A second application of cancellation yields

$$S_{q-z_n} \leq_1 U_n \oplus X.$$

Therefore

$$(4.8) \quad U_n \oplus X \equiv_1 S_{q-z_n}.$$

For $n = 0$, the conclusion

$$U_0 \oplus X \equiv_1 S_q$$

is exactly (4.6). Therefore, for every $n \in \mathbb{N}$, the set $U_n \oplus X$ is a scalar.

It follows that, for every scalar $S_r = mU_n$ and every $X \leq_1 U$, the set $S_r \oplus X$ is a scalar. If $m = 1$, use the preceding conclusion. If $m > 1$, write, up to a computable permutation of the summands,

$$mU_n \oplus X \equiv_1 (m-1)U_n \oplus (U_n \oplus X),$$

and apply scalar addition.

Step 2: finite-rank decomposition of a bounded reduction. Let $B \equiv_{\text{bfo}} U$, and fix a reduction

$$f : B \leq_{\text{bfo}} U$$

whose fibres have size at most some $k \geq 1$. Define

$$(4.9) \quad D_i = \{x : |\{z < x : f(z) = f(x)\}| = i\}, \quad i < k.$$

The sets D_i are computable, form a partition of $\mathbb{N}$, and $f \upharpoonright D_i$ is injective.

We now make a finite nonuniform choice. Fix

$$I = \{i < k : D_i \text{ is infinite}\}$$

and let

$$F = \bigcup_{i \notin I} D_i.$$

Since the D_i form a finite partition of $\mathbb{N}$, the set I is nonempty and F is finite. The set I , the finite set F , and the finitely many membership values of B on F need not be obtained uniformly from an index of f ; they are now fixed and may be hardcoded into the reductions below.

For each $i \in I$, let $d_i : \mathbb{N} \rightarrow D_i$ be the computable increasing bijection and put

$$X_i = \{x : d_i(x) \in B\}.$$

Then $X_i \leq_1 U$, witnessed by $f \circ d_i$. Since $U \leq_{\text{bfo}} B$, Lemma 4.3 shows that B has reservoirs of both colours.

Write

$$I = \{i_0 < i_1 < \cdots < i_{\ell-1}\},$$

where $\ell = |I| \geq 1$, and put

$$J = \bigoplus_{a < \ell} X_{i_a} := \{\langle a, x \rangle_\ell : a < \ell \text{ \& } x \in X_{i_a}\}.$$

Define

$$\pi_J : \mathbb{N} \longrightarrow \mathbb{N} \setminus F$$

by

$$\pi_J(\langle a, x \rangle_\ell) = d_{i_a}(x) \quad (a < \ell, x \in \mathbb{N}).$$

Since the sets D_{i_a} are pairwise disjoint and

$$\mathbb{N} \setminus F = \bigcup_{a < \ell} D_{i_a},$$

the map π_J is a computable bijection from $\mathbb{N}$ onto $\mathbb{N} \setminus F$. Moreover,

$$\begin{aligned} \langle a, x \rangle_\ell \in J &\iff x \in X_{i_a} \\ &\iff d_{i_a}(x) \in B \\ &\iff \pi_J(\langle a, x \rangle_\ell) \in B. \end{aligned}$$

Removing F from the two reservoirs of B and taking their inverse images under π_J gives computable reservoirs of both colours in J . The set

$$D = \mathbb{N} \setminus F$$

is computable, and

$$B \cap \overline{D} = B \cap F$$

is computable from the hardcoded membership values. Apply Lemma 2.3 with

$$A = J, \quad B = B, \quad D = \mathbb{N} \setminus F, \quad \pi = \pi_J.$$

We obtain

$$(4.10) \quad B \equiv_1 J = \bigoplus_{a < \ell} X_{i_a}.$$

Fix $q \in \mathbb{D}_{>0}$. Starting with S_q , apply Step 1 successively to

$$X_{i_0}, \dots, X_{i_{\ell-1}}.$$

Thus $S_q \oplus J$ is a scalar. Since $B \equiv_1 J$, it follows that $S_q \oplus B$ is a scalar as well. Therefore there is a unique $T_B(q) \in \mathbb{D}_{>0}$ such that

$$(4.11) \quad S_q \oplus B \equiv_1 S_{T_B(q)}.$$

Uniqueness follows from Lemma 4.5.

Step 3: the increment is independent of the initial scalar. For $q, r \in \mathbb{D}_{>0}$,

$$\begin{aligned} S_{q+r} \oplus B &\equiv_1 S_q \oplus (S_r \oplus B) \\ &\equiv_1 S_q \oplus S_{T_B(r)} \\ &\equiv_1 S_{q+T_B(r)}. \end{aligned}$$

Thus

$$(4.12) \quad T_B(q+r) = q + T_B(r).$$

Interchanging q and r gives

$$T_B(q) - q = T_B(r) - r.$$

Hence there is a dyadic rational $\delta(B)$ such that

$$(4.13) \quad T_B(q) = q + \delta(B) \quad (q \in \mathbb{D}_{>0}).$$

Since $S_q \leq_1 S_q \oplus B$, scalar calculus gives $T_B(q) \geq q$, and hence $\delta(B) \geq 0$.

The increment is in fact positive. Choose a bounded finite-one reduction $g : U \leq_{\text{bfo}} B$ with fibre bound $h \geq 1$. Define

$$r_g(x) = |\{z < x : g(z) = g(x)\}|.$$

Then $r_g(x) < h$, and the map

$$x \mapsto \langle r_g(x), g(x) \rangle_h$$

is a total computable injective reduction of U to hB . Thus

$$(4.14) \quad U \leq_1 hB.$$

Therefore

$$2U \leq_1 U \oplus hB.$$

We claim that, for every $j \geq 1$ and every $q \in \mathbb{D}_{>0}$,

$$(4.15) \quad S_q \oplus jB \equiv_1 S_{q+j\delta(B)}.$$

For $j = 1$, this follows from (4.11) and (4.13). Suppose that the claim holds for some $j \geq 1$. Then, using associativity of the join, the induction hypothesis, and again (4.11) and (4.13), we have

$$\begin{aligned} S_q \oplus (j+1)B &\equiv_1 (S_q \oplus jB) \oplus B \\ &\equiv_1 S_{q+j\delta(B)} \oplus B \\ &\equiv_1 S_{q+(j+1)\delta(B)}. \end{aligned}$$

Here $q + j\delta(B) \in \mathbb{D}_{>0}$, since $q \in \mathbb{D}_{>0}$ and $\delta(B) \geq 0$ is dyadic. This proves the claim by induction. Taking $q = 1$ and $j = h$, we obtain

$$U \oplus hB \equiv_1 S_{1+h\delta(B)}.$$

Scalar calculus now yields

$$2 \leq 1 + h\delta(B),$$

so

$$(4.16) \quad \delta(B) \geq \frac{1}{h} > 0.$$

Step 4: cancellation recovers B . By (4.13),

$$(4.17) \quad U \oplus B \equiv_1 S_{1+\delta(B)} \equiv_1 U \oplus S_{\delta(B)}.$$

After computably interchanging the two summands on both sides of the forward reduction in (4.17), we obtain

$$B \oplus U \leq_1 S_{\delta(B)} \oplus U.$$

Apply Lemma 4.1 with

$$P = B, \quad Q = S_{\delta(B)}, \quad Z = U.$$

By Lemma 4.4, $U \leq_{\text{bfo}} S_{\delta(B)}$, and hence $U \leq_{\text{fo}} S_{\delta(B)}$. We obtain

$$B \leq_1 S_{\delta(B)}.$$

For the reverse direction, after computably interchanging the summands on both sides of the reverse reduction in (4.17), we obtain

$$S_{\delta(B)} \oplus U \leq_1 B \oplus U.$$

Apply Lemma 4.1 with

$$P = S_{\delta(B)}, \quad Q = B, \quad Z = U.$$

Since $B \equiv_{\text{bfo}} U$, there is a bounded finite-one reduction

$$U \leq_{\text{bfo}} B,$$

which is in particular finite-one. Thus

$$U \leq_{\text{fo}} B,$$

and the lemma gives

$$S_{\delta(B)} \leq_1 B.$$

Thus

$$B \equiv_1 S_{\delta(B)} = mU_n$$

for suitable $m \geq 1$ and n .

Every mU_n is bounded finite-one equivalent to U , so the map

$$\frac{m}{2^n} \mapsto [mU_n]_1$$

is onto the set of one-one degrees in the bounded finite-one degree of U . By Lemma 4.5, it is well defined, injective, and order preserving in both directions. This proves (4.5). $\square$

5. THE DENSE DEGREE

The two preceding theorems now fit together directly. The block construction supplies a set satisfying the hypotheses of the abstract exhaustivity theorem, while the scalar calculus identifies the resulting one-one degrees with the positive dyadic rationals. It remains only to observe that this order is the countable dense linear order without endpoints.

Corollary 5.1. *There exists a bounded finite-one degree such that the one-one degrees contained in it, ordered by $\leq_1$, form a countable dense linear order without endpoints. More precisely, there is a set $U \leq_T \emptyset''$ such that*

$$(\{[B]_1 : B \equiv_{\text{bfo}} U\}, \leq_1) \cong (\mathbb{D}_{>0}, \leq) \cong (\mathbb{Q}, \leq).$$

Proof. Let $U \leq_T \emptyset''$ and $(U_e)_{e \in \mathbb{N}}$ be supplied by Theorem 3.3. Then $(U_e)_{e \in \mathbb{N}}$ is a weak dyadic tower over U , the set U is noncylindrical and has computable reservoirs of both colours, and $(U, (U_e)_{e \in \mathbb{N}})$ has the base absorption property. Hence Theorem 4.6 gives

$$(\{[B]_1 : B \equiv_{\text{bfo}} U\}, \leq_1) \cong (\mathbb{D}_{>0}, \leq).$$

The set $\mathbb{D}_{>0}$ is countable and linearly ordered by the usual order. If $q, r \in \mathbb{D}_{>0}$ and $q < r$, then

$$q < \frac{q+r}{2} < r,$$

and $(q+r)/2 \in \mathbb{D}_{>0}$, so the order is dense. Moreover, for every $q \in \mathbb{D}_{>0}$,

$$0 < \frac{q}{2} < q < 2q,$$

so the order has neither a least nor a greatest element. By Cantor's characterization of the countable dense linear orders without endpoints,

$$(\mathbb{D}_{>0}, \leq) \cong (\mathbb{Q}, \leq).$$

$\square$

Remark 5.2. The preceding corollary combines two deliberately separated parts of the argument. The global construction and its effectivity analysis relative to $\emptyset''$ are confined to Theorem 3.3 and its preparatory lemmas.

By contrast, Theorem 4.6 is an abstract reduction-theoretic argument using only a weak dyadic tower, noncylindricity, computable reservoirs, base absorption, scalar calculus, finite-rank decomposition, and finite-one cancellation. Its proof includes finite nonuniform choices and effective constructions of reductions, but every resulting reduction is an ordinary computable function. Whenever tower witnesses are used, only finitely many levels, fixed for the reduction as a whole, are involved.

No oracle construction or computable uniform family of tower witnesses is required. Nor is a uniform procedure for selecting all the finite nonuniform data asserted. This separation is intended to facilitate independent verification of the proof.

6. CONCLUSION

We have answered Open Question 3 of Richter, Stephan, and Zhang [6] affirmatively. For a noncomputable set $U \leq_T \emptyset''$, the one-one degrees in $[U]_{\text{bfo}}$ are exactly those represented by the finite autojoins mU_n of the levels of a weak dyadic tower, ordered according to the positive dyadic values $m/2^n$. The full internal order is therefore isomorphic to $(\mathbb{Q}, \leq)$, rather than merely containing a copy of it. Together with the m -rigidity obstruction in [1], this shows that the class of sets whose bounded finite-one degree has this exact internal order type is nonempty but null and meager in Cantor space.

A further question suggested by the present construction is whether such an exact dense realization can be obtained with a representative $U \leq_T \emptyset'$. It would also be interesting to develop versions of the absorption-to-exhaustivity principle for other families of representatives, with the aim of obtaining further exact realizations.

ACKNOWLEDGEMENTS

The main result of this paper was discovered by ChatGPT 5.6 Sol Pro (OpenAI).

The author has verified the arguments and bears sole responsibility for the correctness of all statements, proofs, and conclusions.

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